

EULER FUNCTIONS

Generating Function

$$g(x, t) = \frac{2e^{xt}}{e^t + 1} = \sum_{n=0}^{\infty} E_n(t) \frac{t^n}{n!}$$

$$E_0(x) = 1, E_1(x) = x - \frac{1}{2}, E_2(x) = x^2 - x, E_3(x) = x^3 - \frac{3}{2}x^2$$

Identity $\rightarrow E_n(x+1) + E_n(x) = 2x^n$